Kevin Lu

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EDUCATION

Northeastern University Expected April 2026
B.S. in Computer Science & Mathematics, Honors Program GPA: 3.96/4.00

RESEARCH EXPERIENCE

Bau Lab, Interpretable Neural Networks — Northeastern University June 2024 — Present
Undergraduate Researcher

- **Concept Erasure in Diffusion Models (NeurIPS 2025):** Developed framework distinguishing mechanisms in diffusion model unlearning, and designed probing methods to detect residual knowledge (i.e. latent classifier guidance).
- Evaluated 7 erasure methods under 9 probing strategies and showed that concepts appearing erased under standard prompts often persist through alternative pathways, revealing limits of current unlearning methods.

- **Layer-Wise Analysis of Protein Language Models (NEMI Workshop 2025):** Localized binding-relevant information in ESM-2 using difference-of-means representation analysis, linear classifiers, and causal attention-head ablations.
- Found that binding signal peaks in intermediate layers and degrades toward the output; identified late-layer attention heads whose zero-shot ablation improves binding prediction.

- **Mechanistic Interpretability in AI Protein Folding:** Investigating how ESMFold computes secondary structure predictions through gradient attribution and activation patching of intermediate representations. (In Progress)

Takeda Pharmaceuticals June 2025 — Present
Machine Learning Research Intern

- **Distributed Language Model Training**: Finetuned ESM-2 protein language model for masked language modeling over antibody sequences on 8xA100 node, improving CDR region reconstruction accuracy by 32%
- **Sparse Autoencoders for Antibody Optimization**: Trained sparse autoencoders to disentangle biologically interpretable antibody features and applied feature steering during ESM inference, generating over 10,000 high-affinity candidates; 192 variants experimentally validated with 20% increased binding specificity. (Manuscript In Preparation)

Neural Systems Group — Harvard Medical School June 2023 — May 2024
Undergraduate Researcher

- **Signal Processing for Biomedical ML (RISE Expo 2024):** Implemented Kalman filtering algorithms to extract features from ECG waveforms and trained SVR models achieving Grade B AAMI blood pressure prediction.

PUBLICATIONS & PRESENTATIONS

Published Papers

- **Kevin Lu**, Nicky Kriplani, Rohit Gandikota, Minh Pham, David Bau, Chinmay Hegde, & Niv Cohen (2025). *When Are Concepts Erased From Diffusion Models?* In Proceedings of the 39th Conference on Neural Information Processing Systems (NeurIPS 2025).
- **Kevin Lu**, Nicky Kriplani, Rohit Gandikota, Minh Pham, David Bau, Chinmay Hegde, & Niv Cohen (2025) *Where Do Erased Concepts Go?* In CVPR 2025 2nd Workshop on Visual Concepts.

Manuscripts

- Kevin Lu, Shipra Malhotra. *Sparse Autoencoder Feature Steering Enables Targeted Antibody Optimization.* (In preparation).

Additional Presentations

- **New England Mechanistic Interpretability (NEMI) Workshop** August 2025
Kevin Lu, David Bau *To Bind or Not to Bind: A Layer-Wise Dissection of Binding Information in ESM*

- **3rd CVPR Workshop on Generative Models for Computer Vision** June 2025
K. Lu, N. Kriplani, R. Gandikota, M. Pham, D. Bau, C. Hegde, N. Cohen *Where Do Erased Concepts Go?*

- **Northeastern University Research Expo (RISE)** April 2024
Kevin Lu, Ye Yang, Quan Zhang *Predicting Blood Pressure Using AI Models for Physiological Signals*

INDUSTRY EXPERIENCE

Babel Street July 2024 — December 2024
Machine Learning Engineer Intern

- **High-Throughput NLP Pipeline**: Designed multithreaded data streaming service that filters, embeds, and indexes over 1 TB of multilingual news data per week into an Elasticsearch vector database for downstream analysis.
- **Agentic Annotation System**: Developed a multi-agent LLM data annotation system using LangGraph tooling and human-in-the-loop feedback, automating data labeling workflows and reducing annotation costs by 25%.

PROJECTS

Latent Space Exploration for Concept Erasure in Diffusion Models February 2025 — May 2025
Course Project, MATH 7223: Riemannian Optimization

- **Latent Perturbation Optimization**: Investigated recovery of erased concepts in diffusion models via latent-space optimization; used Taylor-series expansion to diagnose gradient plateaus that cause naive optimization to stall.
- **Geometry-Aware Concept Recovery**: Applied Riemannian pullback metrics to identify latent directions of maximal model sensitivity via Jacobian SVD, constraining optimization to semantically meaningful subspaces and improving concept recovery from 7% to 43%.

ConcussionMute June 2024 – December 2024

- **Transformer-Based Drum Separation**: Trained a 700M-parameter transformer model to separate percussive components from music using negative matrix factorization and fourier transform features.
- **Percussive Noise Suppression:** Reduced harsh percussive sounds by 4 dB while preserving melody and harmony, aimed at helping listeners with post-concussion sound sensitivity.

LEADERSHIP & SERVICE

Reviewer, ICLR Main Conference Nov 2025
Invited and served as reviewer for ICLR main track submission on geometric deep learning.
Reviewer, NeurIPS Mechanistic Interpretability Workshop Aug 2025
President, Northeastern Veritas Forum Chapter Sep 2023 – Present
Lead annual forums engaging 150+ students on philosophy, ethics, and faith in everyday college life.
Community Lead, InterVarsity Christian Fellowship Sep 2023 – Present
Teaching Aide, Dr. Martin Luther King Jr. K-8 School Jan 2023 – Apr 2023
Assisted in K-8 STEM program teaching math, reading, and writing skills in 90-minute weekly classes.
Mandarin Elder Service Intern, Action for Boston Community Development Oct 2022 – Jan 2023
Assisted Mandarin-speaking elders in nursing homes and community centers with learning basic technology skills.

AWARDS

Honors Global Support Fund *$6000 grant to conduct research on software infrastructure in Taiwan* December 2025
Khoury Travel Award (2X) *$1000 Travel grant for undergraduate conference presentations* July 2025, October 2025
Veritas Forum Grant (3X) *$1,250 grant for speaker travel, venue costs, and marketing.* September 2023, 2024, 2025
John Martinson Honors Scholarship *Annual $40,000 merit scholarship* September 2022 — May 2026

SKILLS

Programming Languages: Python, Java, C++, MATLAB, Bash, JavaScript
Tools & Frameworks: PyTorch, Pandas, Transformers, LangChain, Docker, BioPython, Accelerate, Scikit, FastAPI
Data & Cloud: HuggingFace, S3, RabbitMQ, ElasticSearch, DynamoDB, Amazon Web Services (AWS)